Probabilistic Machine Learning

Chapter 2: Probability — Univariate Models

Slides based on K. Murphy, *Probabilistic Machine Learning: An Introduction*

Intro to ML
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Outline

Outline

Two interpretations of probability

- **Frequentist**: probability = long-run frequency of a repeatable event. "Fair coin $\Rightarrow$ heads $\approx$ 50% over many flips."
- **Bayesian**: probability = degree of belief / uncertainty, even about one-off events (e.g. "will the polar ice cap melt by 2030?").
- This book adopts the **Bayesian** interpretation — but the mathematical rules of probability are identical either way.

  "Probability theory is nothing but common sense reduced to calculation." — Pierre Laplace, 1812

Types of uncertainty

- **Epistemic / model uncertainty**: ignorance of the true hidden mechanism — reducible with more data.
- **Aleatoric / data uncertainty**: intrinsic stochasticity — irreducible (e.g. a fair coin flip, even with $p = 0.5$ known exactly).

Intuition

Active learning: query points with high $H(p(y \mid x, \mathcal{D}))$. Is this high because of parameter uncertainty $H(p(\theta \mid \mathcal{D}))$ (worth collecting more data) or inherent output noise $H(p(y \mid x, \theta))$ (more data won't help)?

Probability as extension of logic

Definition: Event

A binary proposition A that is true or false (e.g. "it rains tomorrow").

$$0 \leq \Pr(A) \leq 1, \qquad \Pr(\bar{A}) = 1 - \Pr(A)$$

$$\Pr(A, B) = \Pr(A)\Pr(B) \quad \text{if } A, B \text{ independent}$$

$$\Pr(A \vee B) = \Pr(A) + \Pr(B) - \Pr(A, B), \quad = \Pr(A) + \Pr(B) \text{ if mutually exclusive}$$

Conditional probability & independence of events

Key equation

$$\Pr(B \mid A) \triangleq \frac{\Pr(A, B)}{\Pr(A)} \qquad \text{(undefined if } \Pr(A) = 0\text{)}$$

$$A \text{ independent of } B \iff \Pr(A, B) = \Pr(A)\Pr(B)$$

$$A \perp B \mid C \iff \Pr(A, B \mid C) = \Pr(A \mid C)\Pr(B \mid C)$$

Intuition

Variables are rarely *unconditionally* independent, but conditioning on the right intermediate variable C often renders them (conditionally) independent: graph $A - C - B$.

Outline

Random variables

Definition: Random variable (rv)

An unknown/variable quantity X; its set of possible values $\mathcal{X}$ is the **sample space** / state space. An **event** is a subset of outcomes.

E.g. die roll $\mathcal{X} = \{1, \ldots, 6\}$: "seeing a 1" is $X{=}1$, "odd" is $X \in \{1, 3, 5\}$.

Discrete random variables

$\mathcal{X}$ finite/countable $\Rightarrow$ **discrete** rv.

Key equation

Probability mass function (pmf): $p(x) \triangleq \Pr(X = x), \quad 0 \leq p(x) \leq 1, \ \sum_{x \in \mathcal{X}} p(x) = 1$

E.g. uniform pmf $p(x) = 1/4$ on $\{1, 2, 3, 4\}$ vs. degenerate $p(x) = \mathbb{I}[x = 1]$.

Continuous rv's: cdf and pdf

Key equation

Cumulative distribution function: $P(x) \triangleq \Pr(X \leq x)$ (monotone non-decreasing)

$\Pr(a < X \leq b) = P(b) - P(a)$

Probability density function: $p(x) \triangleq \frac{d}{dx} P(x)$

$$\Pr(a < X \leq b) = \int_a^b p(x)\, dx, \qquad \Pr(x < X \leq x + dx) \approx p(x)\, dx$$

Note: $p(x)$ can exceed 1 (it's a density, not a probability).

Quantiles

$P^{-1}(q)$ = value x_q s.t. $\Pr(X \leq x_q) = q$.

- $P^{-1}(0.5)$ = median; $P^{-1}(0.25), P^{-1}(0.75)$ = quartiles.
- Standard normal: central 95% interval is $(\Phi^{-1}(0.025), \Phi^{-1}(0.975)) = (-1.96, 1.96)$.
- General $\mathcal{N}(\mu, \sigma^2)$: 95% interval $\approx (\mu - 1.96\sigma,\ \mu + 1.96\sigma) \approx \mu \pm 2\sigma$.

Joint, marginal, and conditional distributions

Key equation

Sum rule (marginalization): $p(X{=}x) = \sum_y p(X{=}x, Y{=}y)$

Product rule: $p(x, y) = p(x)\, p(y \mid x)$

Chain rule (repeated product rule):

$$p(x_{1:D}) = p(x_1)p(x_2 \mid x_1)p(x_3 \mid x_1, x_2) \cdots p(x_D \mid x_{1:D-1})$$

Builds a high-dimensional joint from conditionals (**graphical models**, Section 3.6).

Independence & conditional independence of RVs

$$X \perp Y \iff p(X, Y) = p(X)p(Y)$$

Mutual independence of $X_1, \ldots, X_n$: joint factorizes for *every* subset.

Key equation

$$X \perp Y \mid Z \iff p(X, Y \mid Z) = p(X \mid Z)\, p(Y \mid Z)$$

Unconditional independence is rare; conditioning on a mediating variable Z (graph $X - Z - Y$) often yields CI — the foundation of graphical models.

Mean and variance

Key equation

$$\mathbb{E}[X] \triangleq \int x\, p(x)\, dx \text{ (or } \textstyle\sum_x x\, p(x)), \qquad \mathbb{E}[aX + b] = a\mathbb{E}[X] + b$$

$$\mathrm{Var}[X] \triangleq \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - \mu^2, \qquad \mathrm{Var}[aX + b] = a^2\mathrm{Var}[X]$$

Properties: X_i: $\mathbb{E}[\sum_i X_i] = \sum_i \mathbb{E}[X_i]$;
For independent: $\mathrm{Var}[\sum_i X_i] = \sum_i \mathrm{Var}[X_i]$; $\mathbb{E}[\prod_i X_i] = \prod_i \mathbb{E}[X_i]$.

$$\mathrm{std}[X] = \sqrt{\mathrm{Var}[X]} = \sigma \quad \text{(same units as } X\text{)}$$

Mode, and conditional moments

$$x^* = \arg\max_x p(x) \qquad \text{(need not be unique if multimodal)}$$

Key equation

Law of total expectation: $\mathbb{E}[X] = \mathbb{E}_Y[\mathbb{E}[X \mid Y]]$

Law of total variance: $\mathrm{Var}[X] = \mathbb{E}_Y[\mathrm{Var}[X \mid Y]] + \mathrm{Var}_Y[\mathbb{E}[X \mid Y]]$

Actually, these are based on the **Law of Total Probability**(willl explain in class).

Example: Lightbulb factories

$\mathbb{E}[X|Y{=}1]{=}5000$, $\mathbb{E}[X|Y{=}2]{=}4000$, $p(Y{=}1){=}0.6 \Rightarrow \mathbb{E}[X] = 5000(.6) + 4000(.4) = 4600.$

Conditional variance: mixture intuition

For a K-component Gaussian mixture (Y=component indicator):

$$\mathrm{Var}[X] = \underbrace{\mathbb{E}_Y[\mathrm{Var}[X \mid Y]]}_{\text{within-component spread}} + \underbrace{\mathrm{Var}_Y[\mathbb{E}[X \mid Y]]}_{\text{between-component spread}}$$

Example:

$\pi_1{=}\pi_2{=}0.5$, $\mu_1{=}0, \mu_2{=}2, \sigma_1{=}\sigma_2{=}0.5$: within-term $= 0.25$, between-term $= 1.0$ — variance dominated by *which* component, not local spread.

Outline

Bayes' rule

Key equation

$$p(H{=}h \mid Y{=}y) = \frac{p(H{=}h)\, p(Y{=}y \mid H{=}h)}{p(Y{=}y)}$$

- $p(H)$: **prior**. $p(Y = y \mid H = h)$ (as a fn. of h): **likelihood**.
- $p(Y = y) = \sum_{h'} p(H{=}h')p(Y{=}y \mid H{=}h')$: **marginal likelihood** (normalizer).
- $p(H \mid Y{=}y)$: **posterior**.

$$\text{posterior} \propto \text{prior} \times \text{likelihood}$$

Using Bayes' rule to update beliefs = **Bayesian inference**.

Example: COVID-19 testing

$H{=}1$: infected. $Y{=}1$: positive test.

- Sensitivity $\Pr(Y{=}1 \mid H{=}1) = 0.875$; specificity $\Pr(Y{=}0 \mid H{=}0) = 0.975$.
- Prior prevalence $p(H{=}1) = 0.1$ (NYC, Spring 2020).

Key equation

$$p(H{=}1 \mid Y{=}1) = \frac{0.875 \times 0.1}{0.875 \times 0.1 + 0.025 \times 0.9} = 0.795$$

$$p(H{=}1 \mid Y{=}0) = \frac{0.125 \times 0.1}{0.125 \times 0.1 + 0.975 \times 0.9} = 0.014$$

COVID example: why this is counter-intuitive

- At prevalence 1%, a positive test only gives 26% chance of infection!
- Intuition (100,000 people, 1000 infected): 875 true positives + 2475 false positives $= 3350$ total positives $\Rightarrow 875/3350 = 0.26$.
- A single positive test is more likely a **false positive** than a true one, when the disease is rare — the base rate matters.

⋆ Inverse problems

⋆ Advanced / optional

Forward: predict outcomes y from world-state h: $p(y \mid h)$.
Inverse: infer h from observed y — often **ill-posed** (many h consistent with the same y), e.g. inferring 3D shape from a 2D image, or speaker intent from ambiguous words.
Bayes' rule + prior $p(h)$ regularizes: $p(h \mid y) \propto p(h)\, p(y \mid h)$.

Outline

Bernoulli distribution

Key equation

$$Y \sim \mathrm{Ber}(\theta): \qquad \mathrm{Ber}(y \mid \theta) \triangleq \theta^y (1-\theta)^{1-y}, \quad p(Y{=}1) = \theta$$

θ denotes probability of *success*

Binomial: number of heads $s = \sum_n \mathbb{I}[y_n = 1]$ in N Bernoulli trials (y_n outcome in the nth toss):

$$\mathrm{Bin}(s \mid N, \theta) \triangleq \binom{N}{s} \theta^s (1-\theta)^{N-s}, \qquad \binom{N}{k} = \frac{N!}{(N-k)!k!}$$

($N{=}1$ recovers Bernoulli.)

The sigmoid (logistic) function

$$p(y \mid x, \theta) = \mathrm{Ber}(y \mid \sigma(f(x; \theta)))$$

Key equation

$$\sigma(a) \triangleq \frac{1}{1 + e^{-a}} \qquad \text{maps } \mathbb{R} \to [0, 1]$$

Key properties:

- $\sigma'(x) = \sigma(x)(1 - \sigma(x))$;
- $1 - \sigma(x) = \sigma(-x)$;
- $a = \mathrm{logit}(p) = \log \frac{p}{1-p}$ (log-odds);
- softplus $\sigma_+(x) = \log(1 + e^x)$, $\sigma'_+(x) = \sigma(x)$.

Binary logistic regression

$$p(y \mid x; \theta) = \mathrm{Ber}(y \mid \sigma(w^T x + b)) = \frac{1}{1 + e^{-(w^T x + b)}}$$

Example: Iris (Virginica vs. not), petal width

Decision boundary at x^* where $p(y{=}1 \mid x^*) = 0.5$ ($x^* \approx 1.7$cm). Confidence grows moving away from the boundary.

Linear regression is inappropriate for classification — its outputs are unbounded, unlike a probability.

Outline

Categorical & multinomial distributions

Key equation

$$\mathrm{Cat}(y \mid \theta) \triangleq \prod_{c=1}^{C} \theta_c^{\mathbb{I}[y=c]} = \prod_{c=1}^{C} \theta_c^{y_c} \text{ (one-hot form)}, \quad \sum_c \theta_c = 1$$

Multinomial: counts $N_c = \sum_n \mathbb{I}[y_n = c]$ over N categorical trials:

$$M(y \mid N, \theta) \triangleq \binom{N}{N_1 \cdots N_C} \prod_{c=1}^{C} \theta_c^{N_c}$$

(N=1 recovers categorical.)

The softmax function

$$p(y \mid x, \theta) = \mathrm{Cat}(y \mid f(x; \theta))$$

Key equation

$$\mathrm{softmax}(a) \triangleq \left[\frac{e^{a_1}}{\sum_{c'} e^{a_{c'}}}, \ldots, \frac{e^{a_C}}{\sum_{c'} e^{a_{c'}}} \right]$$

Temperature a/T: as $T \to 0$, softmax $\to$ argmax ("winner takes all"); as $T \to \infty$, $\to$ uniform.

Multiclass logistic regression

$$p(y \mid x; \theta) = \mathrm{Cat}(y \mid \mathrm{softmax}(Wx + b)), \qquad p(y{=}c \mid x; \theta) = \frac{e^{a_c}}{\sum_{c'} e^{a_{c'}}}$$

- $C = 2$ reduces to binary logistic regression: $\mathrm{softmax}(a)_0 = \sigma(a_0 - a_1)$.
- Multiclass parameterization is over-parameterized relative to binary (2 weight vectors vs. 1) — same predictions, worse interpretability.
- Iris (3-class): decision boundaries are linear in feature space.

The log-sum-exp trick

Naive softmax overflows/underflows for large $|a_c|$ (e.g. $a = (1000, 1001, 1000) \Rightarrow Z = \infty$).

Key equation

$$\log \sum_{c=1}^{C} e^{a_c} = m + \log \sum_{c=1}^{C} e^{a_c - m}, \quad \text{use } m = \max_c a_c$$

$$\mathrm{lse}(a) \triangleq \log \sum_{c} e^{a_c}, \qquad p(y{=}c \mid x) = \exp(a_c - \mathrm{lse}(a))$$

In practice, cross-entropy losses are computed directly from logits a, not from normalized probabilities.

Outline

The Gaussian: cdf

$$\Phi(y;\mu,\sigma^2) \triangleq \int_{-\infty}^{y} \mathcal{N}(z \mid \mu,\sigma^2)\, dz = \frac{1}{2}\left[1 + \operatorname{erf}\left(\frac{z}{\sqrt{2}}\right)\right],\ z = \frac{y-\mu}{\sigma}$$

- μ = mean = mode. σ^2 = variance; **precision** $\lambda = 1/\sigma^2$.
- μ=0, σ=1: **standard normal**.

The Gaussian: pdf

Key equation

$$\mathcal{N}(y \mid \mu, \sigma^2) \triangleq \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y-\mu)^2\right)$$

$\mathbb{E}[\mathcal{N}(\cdot \mid \mu, \sigma^2)] = \mu$; $\mathrm{std} = \sigma$. Note: pdf value can exceed 1 (e.g. $\mathcal{N}(0 \mid 0, 0.1) = 3.99$).

Gaussian regression: homoscedastic vs. heteroscedastic

$$p(y \mid x; \theta) = \mathcal{N}(y \mid f_\mu(x; \theta),\ f_\sigma(x; \theta)^2)$$

- **Homoscedastic** (constant variance): $p(y \mid x; \theta) = \mathcal{N}(y \mid w^T x + b, \sigma^2)$ — ordinary **linear regression** (Ch. 11).
- **Heteroscedastic** (input-dependent variance): $p(y \mid x; \theta) = \mathcal{N}(y \mid w_\mu^T x + b,\ \sigma_+(w_\sigma^T x))$, using softplus to keep variance > 0.
- 95% predictive interval: $[\mu(x) - 2\sigma(x),\ \mu(x) + 2\sigma(x)]$.

Why is the Gaussian so widely used?

1. Only 2, easily-interpreted parameters (mean, variance).
2. **Central limit theorem**: sums of iid rv's $\rightarrow$ Gaussian — great for modeling noise.
3. **Maximum entropy**: fewest assumptions given a fixed mean/variance (Section 3.4.4).
4. Simple, closed-form math $\Rightarrow$ tractable algorithms (Section 3.2).

Intuition

Historical note: "Gaussian" predates Gauss (Laplace, de Moivre). Murphy avoids "normal" — the Gaussian is actually *atypically* well-behaved among distributions.

Dirac delta & truncated Gaussian

$$\lim_{\sigma \to 0} \mathcal{N}(y \mid \mu, \sigma^2) \to \delta(y - \mu), \qquad \int_{-\infty}^{\infty} \delta(x)\, dx = 1$$

Sifting property: $\int f(y)\delta(x - y)\, dy = f(x)$.

Key equation

Truncated Gaussian on (a, b):

$$\mathcal{N}(x \mid \mu, \sigma^2, a, b) = \frac{\frac{1}{\sigma}\phi(\frac{x-\mu}{\sigma})}{\Phi(\frac{b-\mu}{\sigma}) - \Phi(\frac{a-\mu}{\sigma})} \mathbb{I}[a < x < b]$$

⋆ Empirical distribution

⋆ Advanced / optional

Given samples $\mathcal{D} = \{x^{(1)}, \ldots, x^{(N)}\}$, approximate the pdf with delta spikes:

$$\hat{p}_N(x) = \frac{1}{N} \sum_{n=1}^{N} \delta_{x^{(n)}}(x), \qquad \hat{P}_N(x) = \frac{1}{N} \sum_{n=1}^{N} u_{x^{(n)}}(x)$$

The cdf is a "staircase" / "step" function with jumps of $1/N$ at each sample.

Outline

⋆ Transformations: discrete & continuous

⋆ Advanced / optional

Discrete: $p_y(y) = \sum_{x:f(x)=y} p_x(x)$.

Continuous (via cdf's): $P_y(y) = \Pr(f(X) \leq y)$. If f invertible: differentiate to get the pdf; if not, use numerical integration or Monte Carlo.

⋆ Change of variables

⋆ Advanced / optional

Scalar, $y = f(x)$, $x = g(y)$:

$$p_y(y) = p_x(g(y)) \left| \frac{d}{dy} g(y) \right|$$

Multivariate, $y = f(x)$, invertible, inverse g:

$$p_y(y) = p_x(g(y)) \, |\det J_g(y)|$$

E.g. polar coordinates $x = (r\cos\theta, r\sin\theta)$: $|\det J_g| = r \Rightarrow p_{r,\theta}(r,\theta) = p_{x_1,x_2}(r\cos\theta, r\sin\theta)\, r$.

⋆ Moments of a linear transformation

⋆ Advanced / optional

For $y = Ax + b$:

$$\mathbb{E}[y] = A\mu + b, \qquad \mathrm{Cov}[y] = A\Sigma A^T$$

Scalar case $y = a^T x + b$: $\mathrm{Var}[y] = a^T \Sigma a$; e.g. $\mathrm{Var}[x_1 + x_2] = \Sigma_{11} + \Sigma_{22} + 2\Sigma_{12}$.

⋆ Convolution theorem

⋆ Advanced / optional

For independent x_1, x_2, $y = x_1 + x_2$:

$$p(y) = p_1 \circledast p_2 = \int p_1(x_1)\, p_2(y - x_1)\, dx_1 \qquad \text{(“flip and drag”)}$$

Special case — **sum of two Gaussians is Gaussian**:

$$\mathcal{N}(\mu_1, \sigma_1^2) \circledast \mathcal{N}(\mu_2, \sigma_2^2) = \mathcal{N}(\mu_1 + \mu_2,\ \sigma_1^2 + \sigma_2^2)$$

E.g. sum of two dice rolls: triangular pmf on $\{2, \ldots, 12\}$, already looking bell-shaped.

⋆ Central limit theorem

⋆ Advanced / optional

$X_1, \ldots, X_N$ iid, mean μ, variance σ^2; $S_N = \sum_n X_n$. Then

$$Z_N \triangleq \frac{S_N - N\mu}{\sigma\sqrt{N}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{N}} \xrightarrow{d} \mathcal{N}(0,1)$$

Explains why sums/averages of many small, independent effects look Gaussian — and why the Gaussian is such a good noise model.

⋆ Monte Carlo approximation

⋆ Advanced / optional

When $p(y)$ for $y = f(x)$ is analytically intractable: draw many samples $x^{(s)} \sim p(x)$, compute $y^{(s)} = f(x^{(s)})$, approximate

$$p_S(y) \approx \frac{1}{S} \sum_{s=1}^{S} \delta(y - y^{(s)})$$

Simple, general, and widely used throughout the book (and beyond) for approximate inference.

Summary

- Probability as extended logic: sum rule, product rule, chain rule, (conditional) independence.
- Random variables: pmf/cdf/pdf, moments, and their conditional analogues.
- **Bayes' rule** $\Rightarrow$ prior $\times$ likelihood $\rightarrow$ posterior; base-rate effects can be counter-intuitive.
- Key distributions: Bernoulli/Binomial, Categorical/Multinomial (+ sigmoid/softmax), Gaussian, and heavy-tailed/positive-support alternatives (Student-t, Laplace, Beta, Gamma).
- Transformations of rv's, convolution, the CLT, and Monte Carlo approximation.